

HAZOP Study Report

Crude Distillation Unit — Node 4, Column Overhead and Charge Pump Circuit — Report HZ-CDU-04 Rev. 2

SYNTHETIC DOCUMENT - created for a Smart India Hackathon prototype demonstration. Not real plant data. No sponsor information is used.

1. Distribution

This report is classified RESTRICTED. Distribution is limited to the process safety group, the unit manager, and nominated members of the operations team. It shall not be issued to contractors or third parties without written authorisation from the head of process safety.

2. Safety Instrumented Functions

Tag	Function	Set point	SIL
PSHH-401	Column overhead high pressure trip	4.2 bar g	SIL 2
	Deadband	0.3 bar	
LSLL-412	Column bottom low level trip	18 percent	SIL 2
TSHH-430	Furnace outlet high temperature trip	372 deg C	SIL 3
FSLL-104	P-101 low flow trip	95 m3/h	SIL 1

The interlock set point for the crude distillation column is 4.2 bar with a 0.3 bar deadband. The trip initiates overhead vapour routing to flare and simultaneously trips the charge pump P-101. Override of this interlock requires written authorisation from the unit manager and is permitted only during commissioning.

3. Selected Deviations

Node	Deviation	Cause	Consequence	Safeguard
4.1	High pressure	Overhead condenser fouling	Column relief lift, product loss	PSHH-401 at 4.2 bar, PSV-403
4.2	No flow	P-101 seal failure	Furnace tube overheating	FSL-104 trip, standby P-101A
4.3	Low level	Bottoms pump runaway	Pump cavitation, vapour breakthrough	LSLL-112 trip
4.4	High temperature	Burner control failure	Tube rupture, fire	TSHH-430 at 372 deg C

4. Actions Arising

Action	Description	Owner	Due
A-04-01	Confirm PSHH-401 proof test interval is 12 months	Process safety	2026-09-30
A-04-02	Record flush differential pressure on P-101 each shift	Operations	2026-08-15
A-04-03	Review seal failure trend against flush filter changes	Reliability	2026-10-31